

20260928: gLike

Overview

Given the *time-based* performance improvements of CMA-ES over gLike (natively), I tried some tests with varying the value of κ .

CMA-ES Update: Normalization

I figured out and implemented a normalization scheme on a per-parameter basis. I approximate a parameter scaling factor such that the normalized parameter should be between 0 and 1, thus a precision of 0.05 represents about a 5% wiggle-room.

As is, parameter scaling is encoded with the model, but if no parameter scaling is provided, then:

```
if parameter_scales is None:
    parameter_scales = {}

    for name in x0:
        if name.startswith("N_"):
            parameter_scales[name] = 50000
        elif name.startswith("r"):
            parameter_scales[name] = 1.0
        elif name.startswith("m"):
            parameter_scales[name] = 1.0
        elif name.startswith("gr"):
            parameter_scales[name] = 1.0

search = Search(
    x0,
    bounds=bounds,
    precision=precision,
    parameter_scales=parameter_scales
)
```

```

==> CMA_kappa_25000_3G09_12401992_10.log <==
Generation    5    Likelihood = -107293.166932
{'t1': 8.334510437094382, 't2': 59.68465267922789, 't3': 1708.575814158042, 'N_anc': 1205}

Final normalized parameters:
{'t1': 0.17009204973662004, 't2': 0.6631628075469765, 't3': 0.1717161622269389, 'N_anc': 1205}

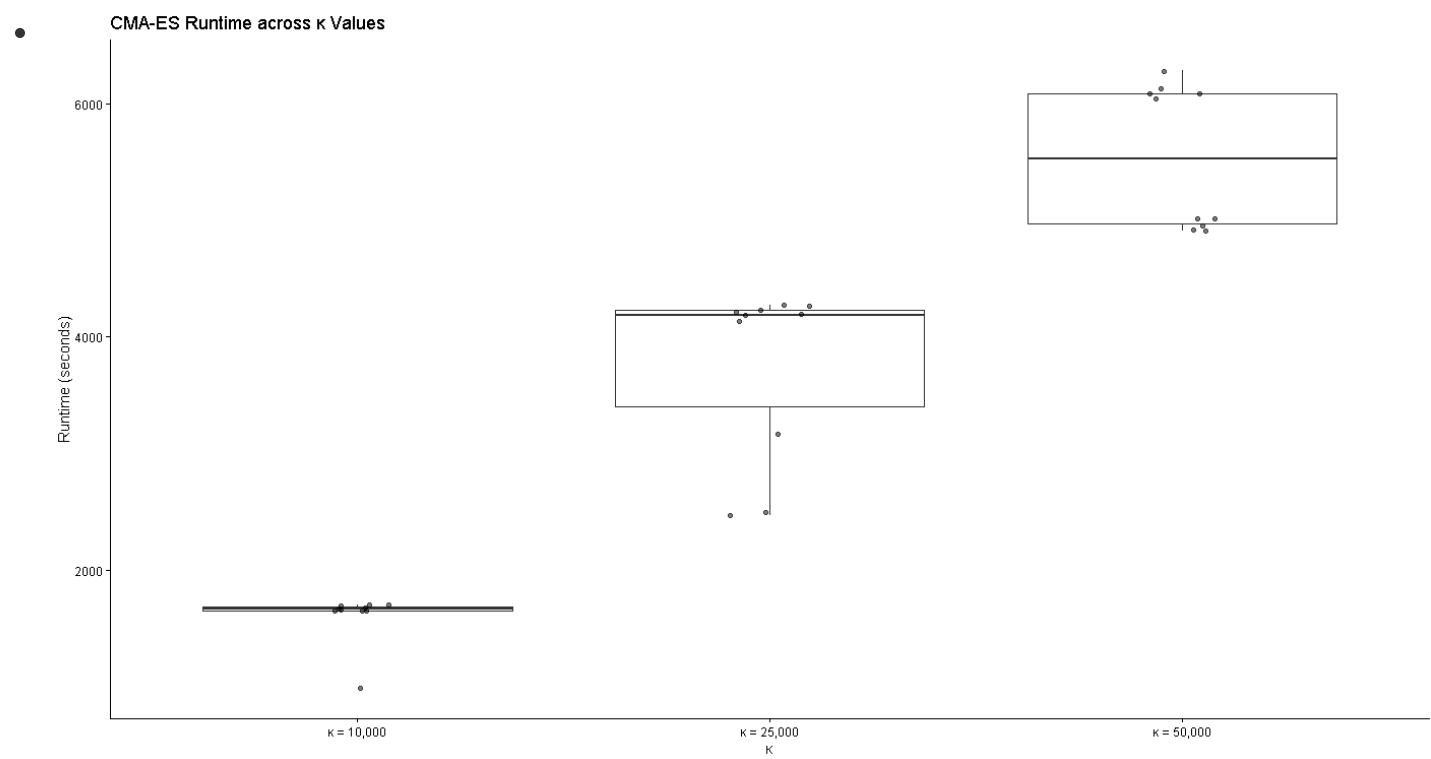
Final demographic parameters:
{'t1': 8.334510437094382, 't2': 59.68465267922789, 't3': 1708.575814158042, 'N_anc': 1205}

```

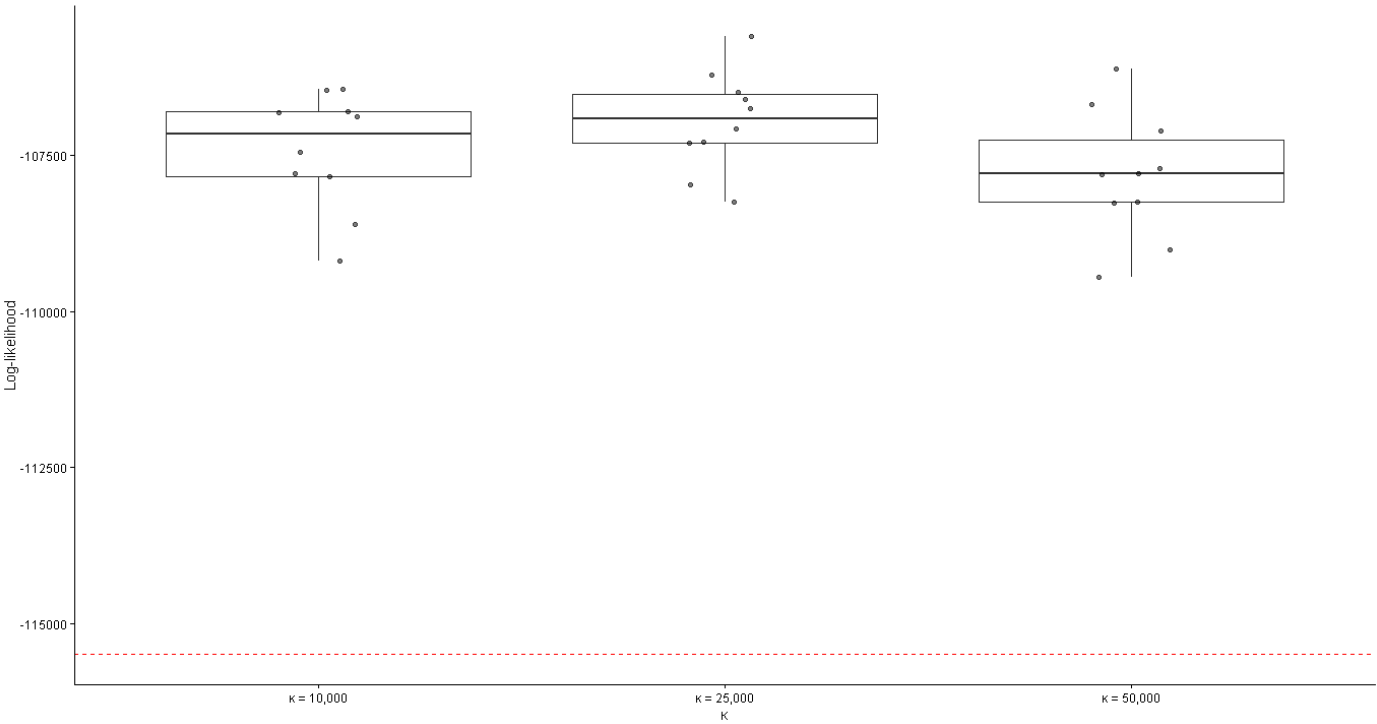
CMA-ES Kappa results comparison

Increasing `kappa` substantially icnreases runtime, but doesn't offer substantial improvement in model fit. Quite annoyingly, I'm still finding cases where the logp of the best model is better fit than the actual true log likelihood. Quite annoyingly, there's still not that much movement in the parameters, despite normalization of scaling. Anecdotally, There are still only a maximum of 5 generations being brun using CMA-ES. I'm looking into tweaking this, but I think it's internal to the `cma` python package that I imported.

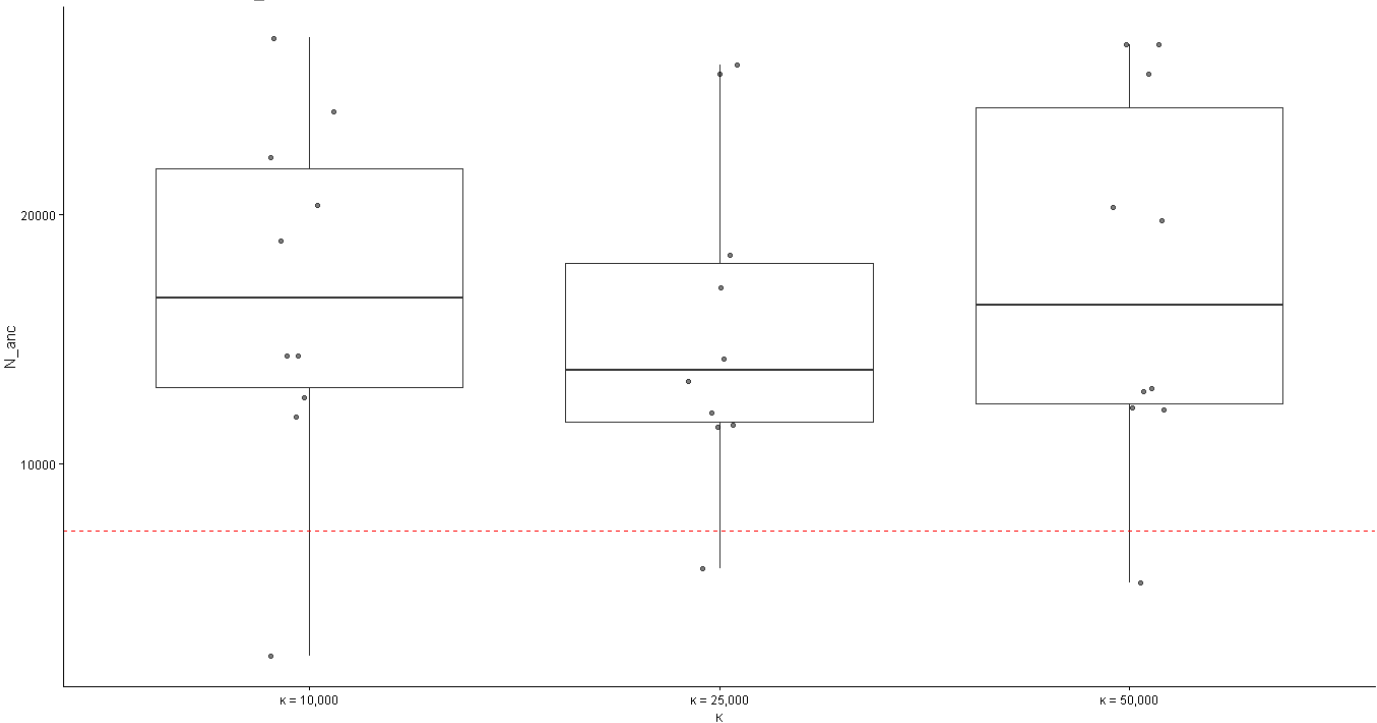
True values are shown as a dashed red line.



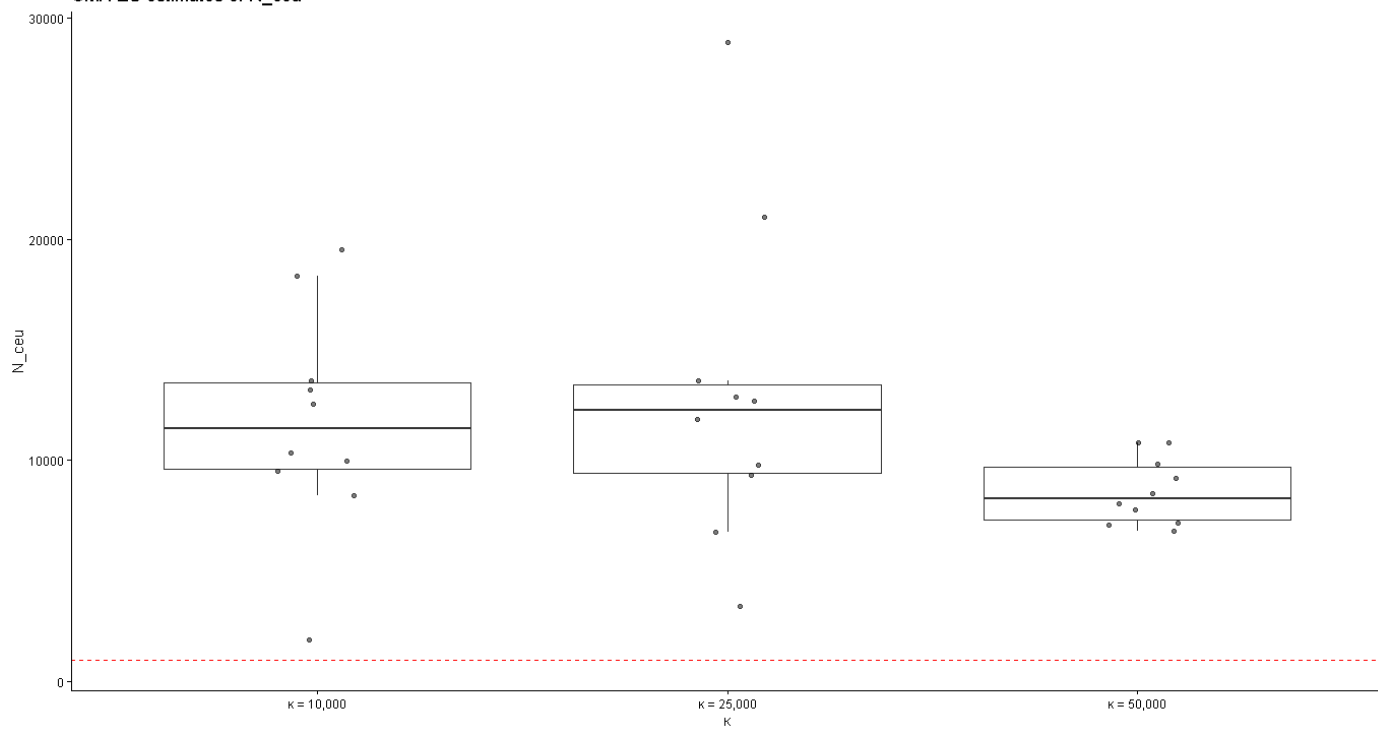
CMA-ES Log-likelihood across κ Values



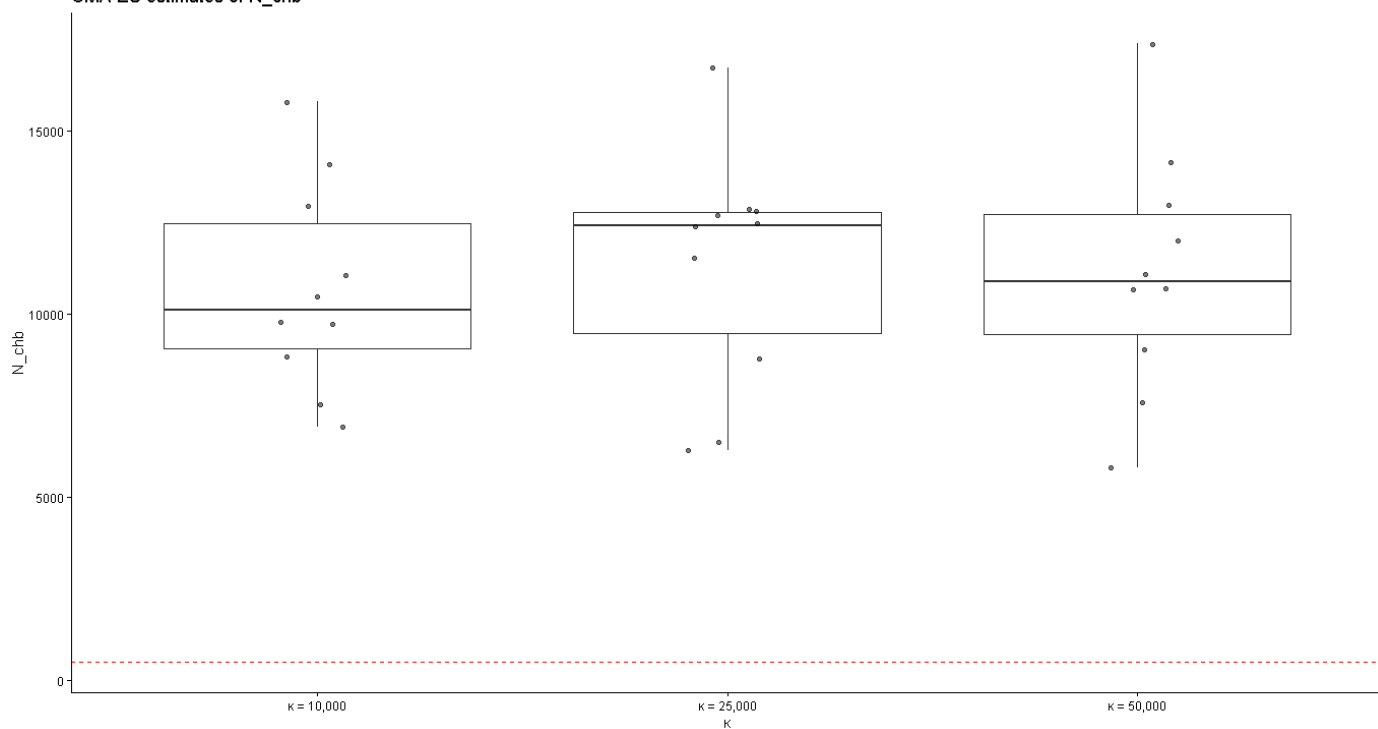
CMA-ES estimates of N_{anc}



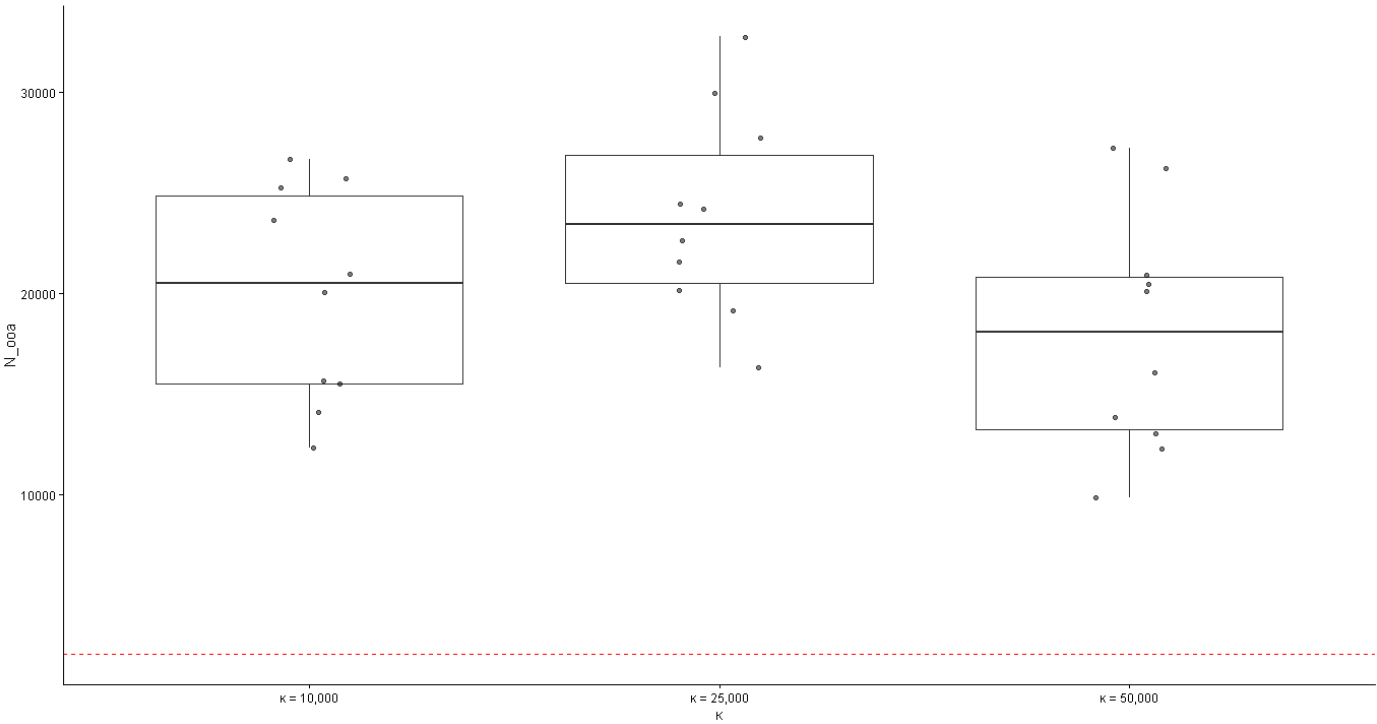
CMA-ES estimates of N_ceu



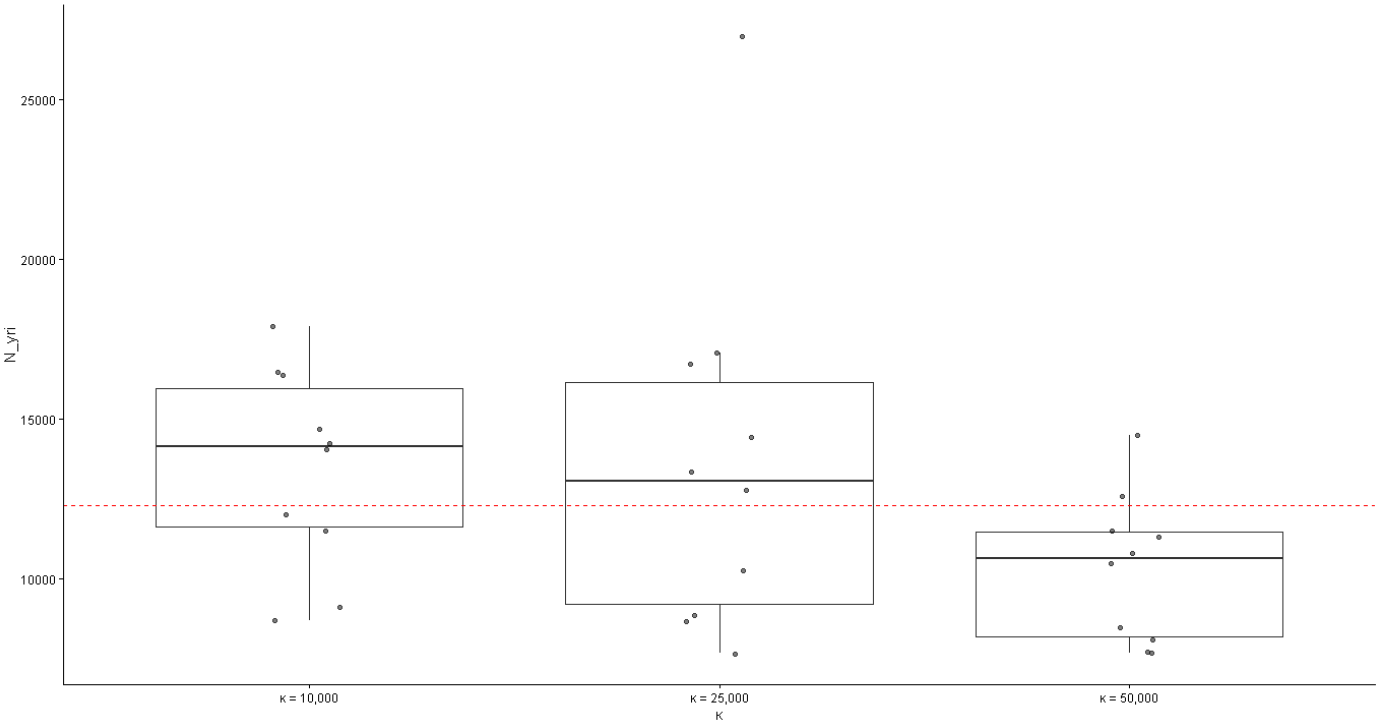
CMA-ES estimates of N_chb



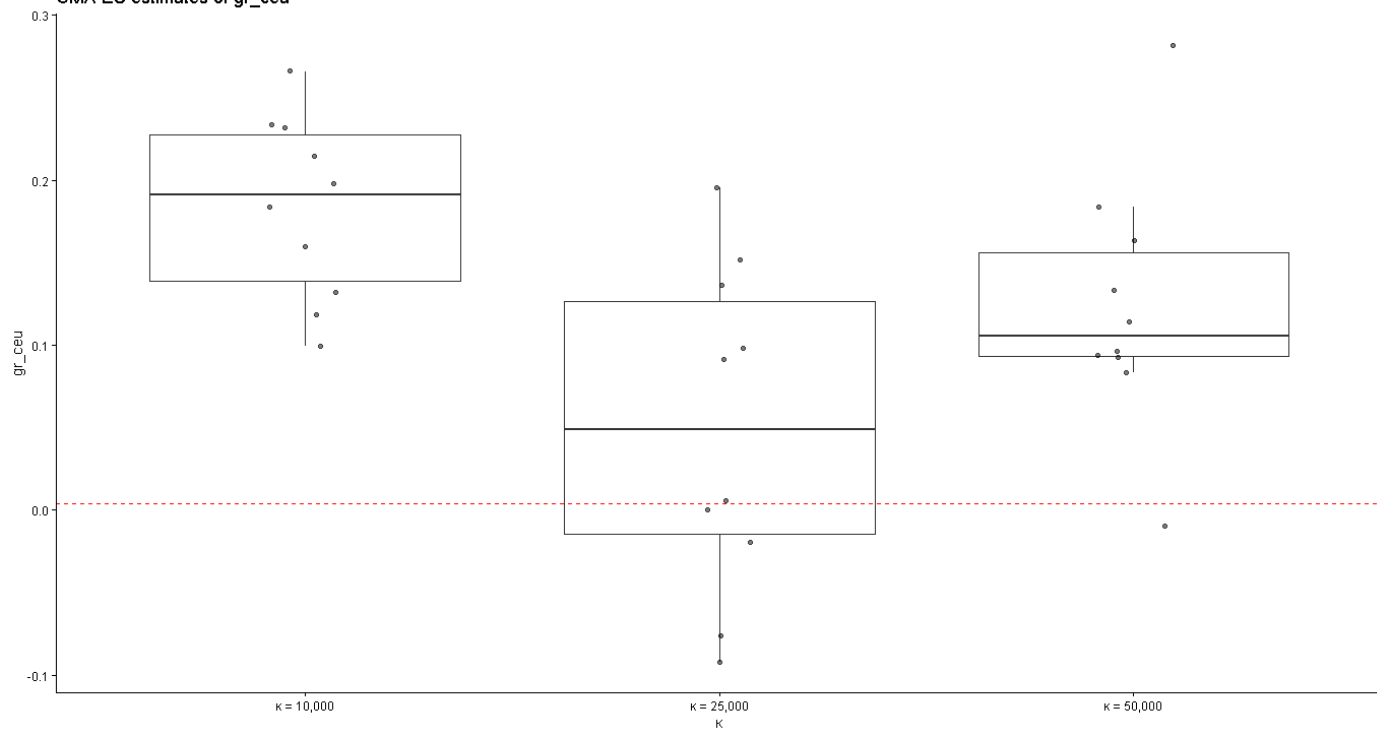
CMA-ES estimates of N_ooa



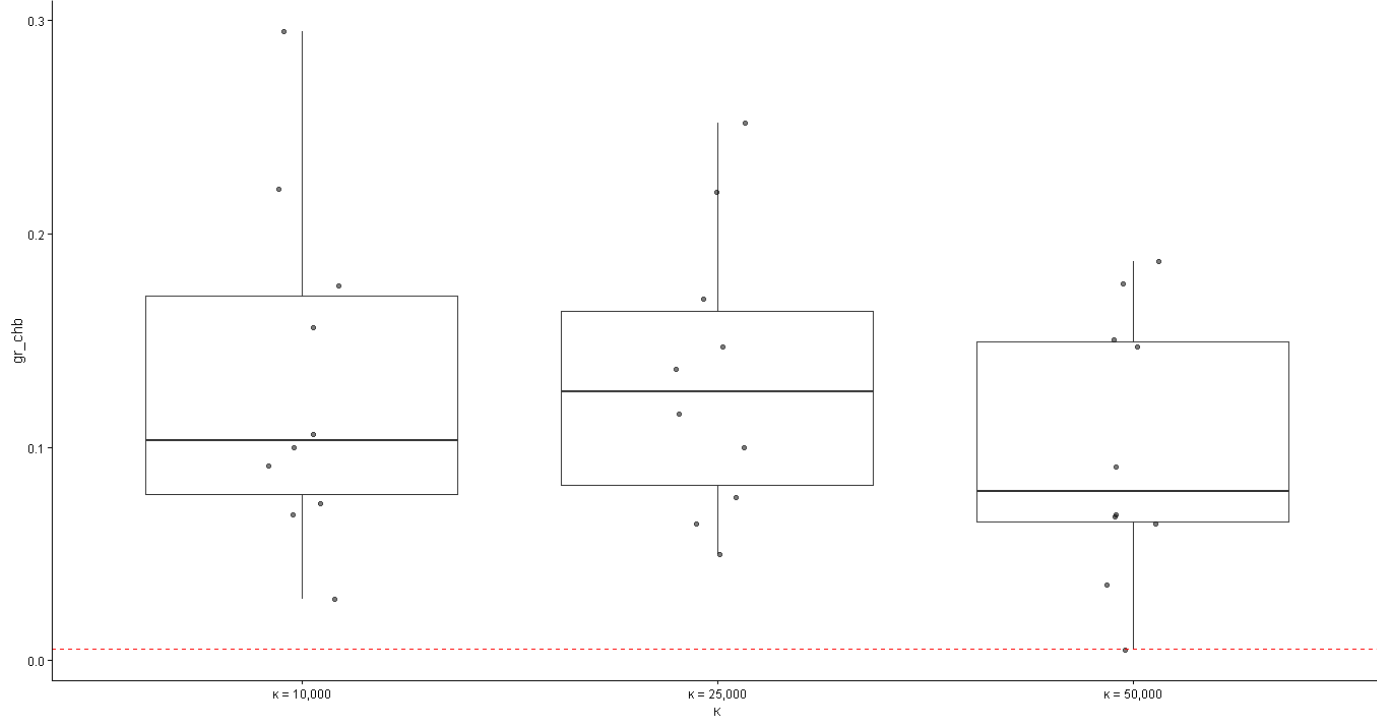
CMA-ES estimates of N_yri



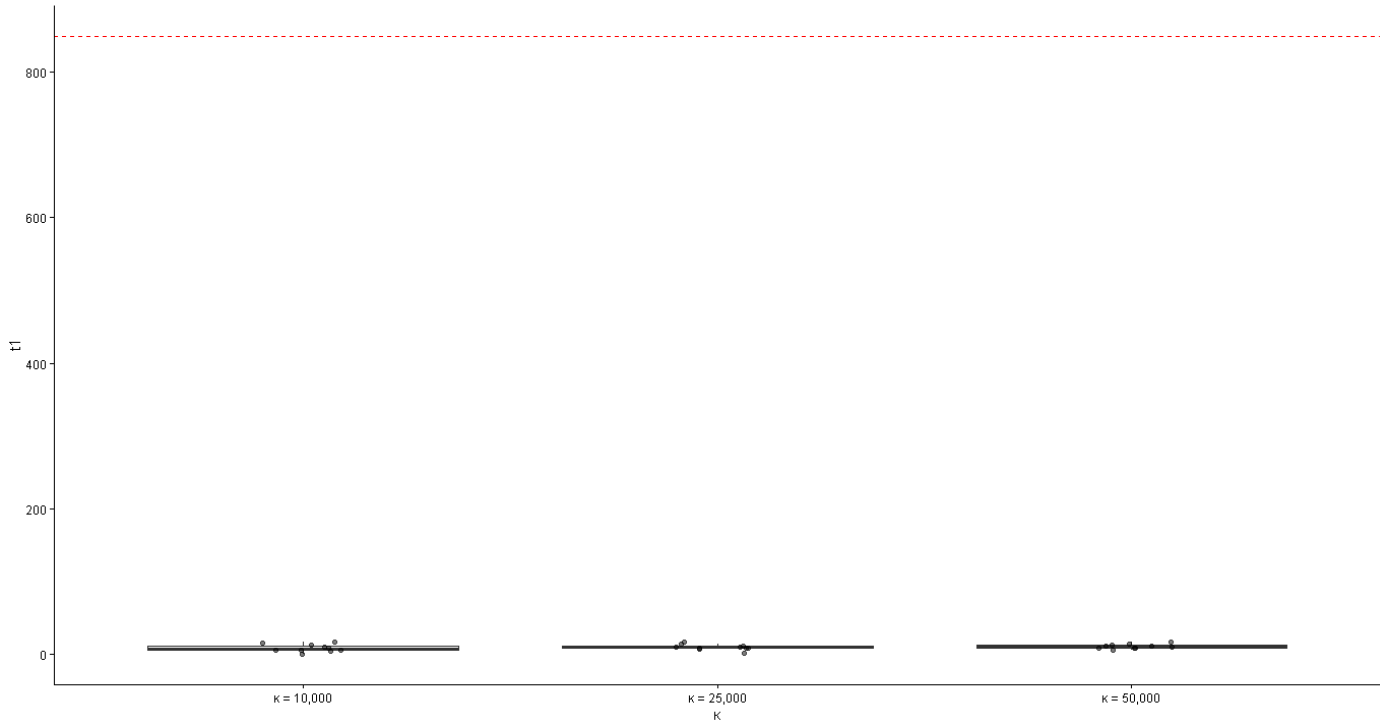
CMA-ES estimates of gr_ceu



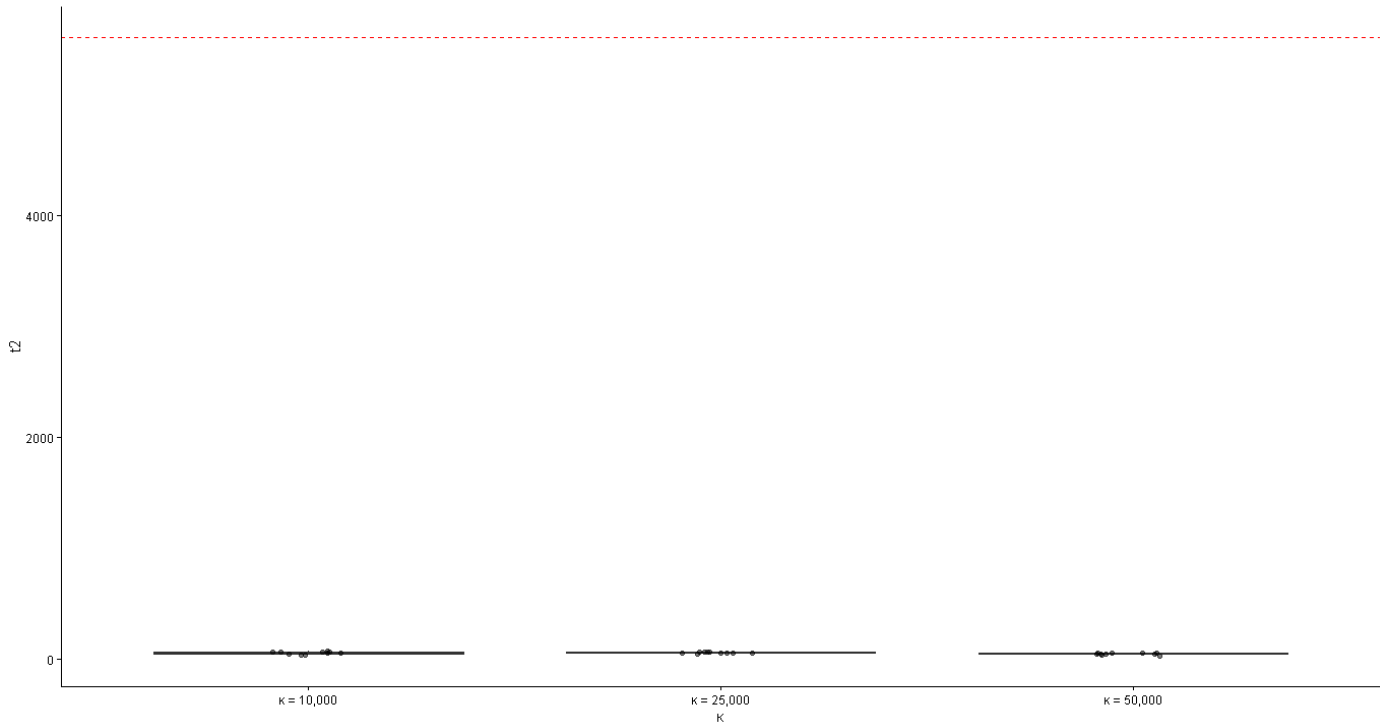
CMA-ES estimates of gr_chb



CMA-ES estimates of t1



CMA-ES estimates of t2



CMA-ES estimates of t_3

